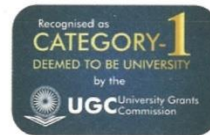




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**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING  
SCHOOL OF COMPUTING  
10214CS601- MINOR PROJECT – I  
ACADEMIC YEAR – 2025-2026(SUMMER SEMESTER)**

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# AI BASED SKIN LESION CLASSIFIER

## **AGENDA**

- **Introduction**
- **Literature Survey**
- **Problem Statement**
- **Proposed Methodology**
- **Proposed System Architecture**
- **Implementation Plan/Timeline**

# Introduction

- Skin cancer is often hard to detect early. This project uses deep learning with explainable AI to classify skin lesions accurately and help doctors understand the model's decisions.
- To build a deep learning model for classifying different types of skin lesions
- To improve early detection and support medical diagnosis with accurate results

# Introduction

- To design and implement a web-based/classifier tool that can detect and classify skin lesions (benign vs malignant).
- To improve accuracy using image preprocessing + CNN-based deep learning model.
- To provide a user-friendly interface with secure login and image upload features.

# Introduction

- Skin cancer is increasing worldwide, and early detection is very important.
- Our project uses Convolutional Neural Network (CNN) in Python to automatically classify skin lesions as benign or malignant.
- A simple web interface is provided so users can upload images and get instant results.
- The system is designed to be fast, easy to use, and helpful for early diagnosis.

# Literature Survey

- Existing Research: Some deep learning models like VGG16, ResNet, and Inception are used to classify skin lesions.
- Datasets: The ISIC dataset is the most common collection of skin images for training these models.
- Findings: These models give good accuracy but face problems like unbalanced data (more images of one type than another) and variations in lighting or skin color.
- Gap: There is still a need for a system that is simple to use, works in real-time, and can be easily used by people (not just researchers).

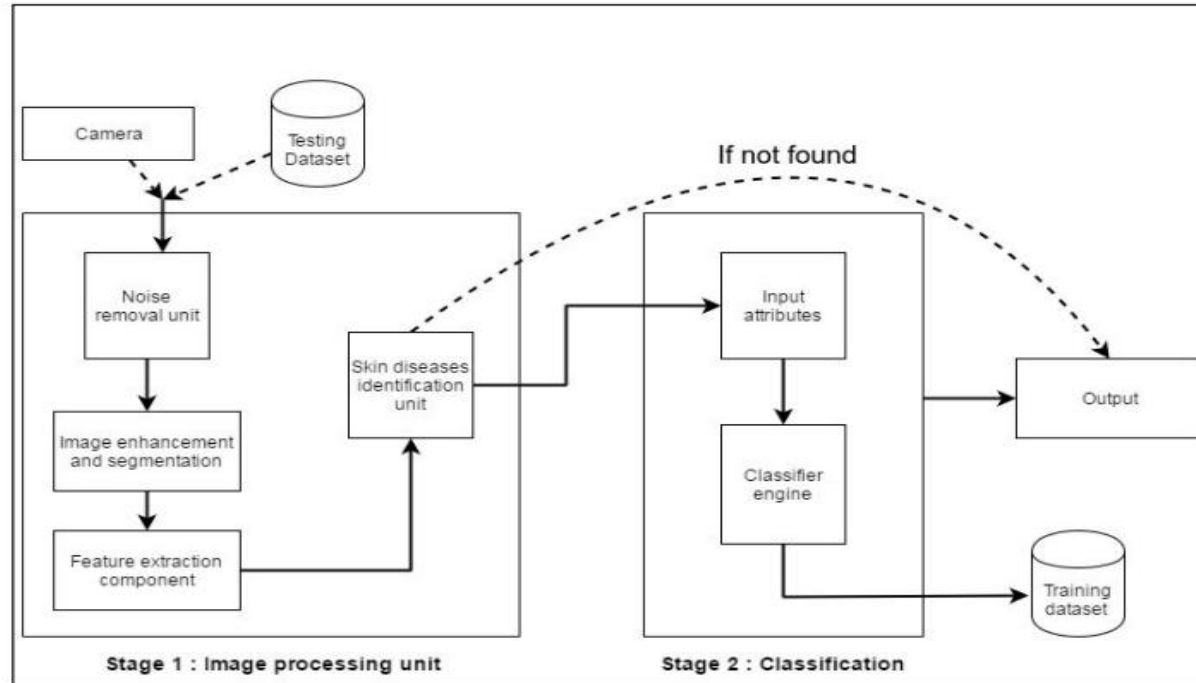
# Problem Statement

- Lack of accessible diagnostic tools for early detection of skin cancer.
- Dependence on dermatologists → delays and higher costs.
- Misdiagnosis possible due to human error.
- An automated, accurate, and fast system to classify lesions is done

# Proposed Methodology

- **Dataset Collection** – ISIC (International Skin Imaging Collaboration) dataset of dermoscopic images.
- **Image Preprocessing** – Resizing, noise removal, grayscale, edge detection.
- **Model Development** – CNN-based deep learning classifier.
- **Evaluation** – Status, Accuracy , Treatment.
- **Deployment** – Flask web app with login/registration.

# Proposed System Architecture



# Summary of Module -1

- Secure login and registration system.
- Only authorized users can access classification service.
- Basic session handling to keep users logged in safely.

# Summary of Module -2

- Upload lesion images via web interface.
- Preprocessing steps applied:
  - i. Grayscale conversion
  - ii. Sharpening & thresholding
  - iii. Edge detection
- Prepares data for accurate model prediction.

# Summary of Modules -3

- Convolutional Neural Network trained on ISIC dataset.
- Extracts features such as texture, color, and shape.
- Classifies lesions into Benign or Malignant.
- Provides probability score for prediction confidence.

# Summary of Modules -4

- Predicts the type of skin disease from the uploaded image.
- Shows confidence level and accuracy of the prediction.
- Suggests possible treatment or next steps.
- Users can save or download their test results locally.

# Software & Hardware Requirements Specification

- Windows 10/11 or Linux
- Python 3.x
- IDE: Visual Studio Code / PyCharm / Jupyter Notebook
- Libraries: TensorFlow, OpenCV, NumPy, Pandas, Matplotlib
- Front-end: HTML, CSS, JavaScript
- Web framework: Flask (if using web interface)

# Implementation Plan/Timeline

## **Key Milestones:**

- **Week 1-2:** Dataset collection & preprocessing.
- **Week 3-4:** Model training & optimization.
- **Week 5-6:** Web app development & model integration.
- **Week 7-8:** Testing, evaluation, documentation, final presentation.

# THANK YOU